A BOUNDED ABOVE COMPLEX OF FLAT MODULES ON A RINGED SITE IS K-FLAT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument, including the colimit reduction and the vanishing of the E_1 terms; the author's closing phrase is retained verbatim.

Editorial note. Cross-chapter restatement: the same label occurs in chapter cohomology for a ringed space. That statement is the special case of this ringed-site statement, so per the lane-s4 rule only this more general occurrence is counted here.

Editorial note. The single labelled reference occurring inside the counted spans points to another chapter of the Project; it is carried in assembly.external_references as a non-load-bearing navigation pointer recorded at the invoking line.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- K-flat complex of O-modules. Same chapter, the definition of K-flatness: the total tensor product with every acyclic complex is acyclic. Supplies the property proved for a bounded above complex of flat modules.
- Flat module on a ringed site and exactness of tensorisation. Same chapter, the definition of flatness. Supplies that each K^q is flat, which makes the E_1 terms of the spectral sequence vanish.
- Truncations tau_{<= m} and their behaviour under tensor products. Same chapter, the truncation functors and the termwise identity L^bullet = colim_m tau_{<= m}L^bullet. Supplies the reduction from an arbitrary acyclic complex to a bounded above one, using that filtered colimits are exact.

Source reference. Original lemma 0132: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3142–3142. Cross-chapter navigation pointer invoked at proof lines 3141-3142, where the author invokes the first quadrant spectral sequence of a double complex. The target chapter is present in this repository only as another lane's snapshot, so the pointer records the invoking line; it only resolves the reference for the reader, and no bibliography citation is involved.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. A bounded above complex of flat $\mathcal{O}$-modules is K-flat.*

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Proof. Let $\mathcal{K}^\bullet$ be a bounded above complex of flat $\mathcal{O}$-modules. Let $\mathcal{L}^\bullet$ be an acyclic complex of $\mathcal{O}$-modules. Note that $\mathcal{L}^\bullet = \operatorname{colim}_m \tau_{\leq m}\mathcal{L}^\bullet$ where we take termwise colimits. Hence also

$$\operatorname{Tot}(\mathcal{K}^\bullet \otimes_{\mathcal{O}} \mathcal{L}^\bullet) = \operatorname{colim}_m \operatorname{Tot}(\mathcal{K}^\bullet \otimes_{\mathcal{O}} \tau_{\leq m}\mathcal{L}^\bullet)$$

termwise. Hence to prove the complex on the left is acyclic it suffices to show each of the complexes on the right is acyclic. Since $\tau_{\leq m}\mathcal{L}^\bullet$ is acyclic this reduces us to the case where $\mathcal{L}^\bullet$ is bounded above. In this case the spectral sequence of Homology, Lemma 0132 has

$${}'E_1^{p,q} = H^p(\mathcal{L}^\bullet \otimes_R \mathcal{K}^q)$$

which is zero as $\mathcal{K}^q$ is flat and $\mathcal{L}^\bullet$ acyclic. Hence we win. □